02LQD Specification and Simulation of Digital Systems

2018/02/09

1. Combinational circuit design (5 points)

Thermometer-to-binary (TM2B) encoders are important components of Flash Analog-to-Digital Converters (FADC). The thermometer outputs a pure binary signal X on 7 bits. The encoder receives X and generates Z on 3 bits according to the following table:

X	Z
0000000	000
0000001	001
0000011	010
0000111	011
0001111	100
0011111	101
0111111	110
1111111	111

Design in VHDL an encoder implementing the above specification according to the following description styles:

1. behavioural using IF THEN ELSEIF statements
2. behavioural using the CASE WHEN statement.

2. FSM design (10 points)

In digital communication, a special synchronization pattern, known as a ***preamble,*** is used to indicate the beginning of a packet. For example, the Ethernet II preamble includes eight repeating octets of "10101010". Design a synchronous circuit that generates the "10101010" pattern. The circuit has a 1 bit input signal, `start`, and a 1 bit output signal, `data_out`. When `start` is '1', the "10101010" output will be generated in the next eight clock cycles. Reset is synchronous.

1. Draw the state diagram (**2 points**)
2. Design the Moore FSM resorting to the VHDL behavioural description style with 2 processes (**3 points**)
3. Design the HLSM (**3 points**).

3. Memory-based design (15 points)

Given a 2^3 x 4 ROM, design a system that counts how many ROM values are at 0. It should start counting when a *Go* signal is asserted and it should set a *Finish* signal to 1 when counting is over and the result is available on *Counter*. The system acts as top-level entity and its structure is composed of a ROM and of an interface module. The interface module is a FSM that reads data sequentially from the ROM, compares them to 0 and updates the count accordingly. Reset is synchronous. Define the signals exchanged by the interface module and the ROM. Create a suitable testbench.

Exam rules:upload on Portale della didattica in section Elaborati:

- running VHDL code and testbench
- short report (max 3 pages) describing the solution highlighting differences w.r.t. the solution handed in at the written exam

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02LQD Specification and Simulation of Digital Systems

2018/02/23

1. Combinational circuit design (5 points)

Excess-3 is a self-complementary, biased and redundant code. Each decimal digit is expressed on 4 bits. The following table matches BCD and excess-3:

BCD	excess-3
0000	0011
0001	0100
0010	0101
0011	0110
0100	0111
0101	1000
0110	1001
0111	1010
1000	1011
1001	1100

Design in VHDL a BCD to excess-3 transcoder implementing the above table according to the following description styles:

1. dataflow using conditional signal assignments (WHEN ELSE statements)
2. dataflow using selected signal assignments (WITH SELECT WHEN statements)

2. FSM-D design (25 points)

Fibonacci numbers are recursively defined as follows:

$FIB_n = FIB_{n-2} + FIB_{n-1}$ $\quad\quad\quad\quad$ $n>1$

$FIB_0 = 0$

$FIB_1 = 1$

Design a FSM-D to **iteratively** compute Fibonacci numbers of order `n`. Note that `n` is on 4 bits. Note that, being $FIB_{15}=610$, the output `data_out` is on 10 bits. Computation starts when a `start` signal is asserted for one clock cycle. When computation is over the `ready` signal is asserted for one clock cycle. Reset is asynchronous. The output `data_out` makes sense only when *`ready`* is 1.

Steps:

1. Draw the HLSM state diagram and write the VHDL code to capture the HLSM behavior (**10 points**)
2. Design a FSM-D (**15 points**):
   - create 2 processes for behavioural datapath description from the HLSM state diagram
   - create 2 processes for behavioural controller description from the HLSM state diagram
   - create a process for behavioral output description from the HLSM state diagram.

Exam rules:

- upload on Portale della didattica in section Elaborati:
  - running VHDL code and testbench
  - short report (max 3 pages) describing the solution highlighting differences w.r.t. the solution handed in at the written exam

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02LQD Specification and Simulation of Digital Systems

2018/07/02

1. Combinational circuit design (5 points)

8 4 -2 -1 is a self-complementary, biased and redundant code. Each decimal digit is expressed on 4 bits. The following table matches BCD and 8-4-2-1:

BCD	8 4 -2 -1
0000	0000
0001	0111
0010	0110
0011	0101
0100	0100
0101	1011
0110	1010
0111	1001
1000	1000
1001	1111

Design in VHDL a BCD to 8-4-2-1 transcoder implementing the above table according to the following description styles:

1. dataflow using conditional signal assignments (WHEN ELSE statements)
2. dataflow using selected signal assignments (WITH SELECT WHEN statements)

2. FSM-D design (25 points)

The factorial of a positive integer is recursively defined as follows:

fact(n) = n * fact(n-1) n>1
fact(0) = 1

Design a FSM-D to **iteratively** compute the factorial of a positive integer `n` expressed on 3 bits. Note that, being fact(7)=5040, the output `data_out` is on 13 bits. Computation starts when a *`start`* signal is asserted for one clock cycle. When computation is over the *`ready`* signal is asserted for one clock cycle. Reset is asynchronous. The output `data_out` makes sense only when *`ready`* is 1.

Steps:

1. Draw the HLSM state diagram and write the VHDL code to capture the HLSM behavior (**10 points**)
2. Design a FSM-D (**15 points**):
   - create 2 processes for behavioural datapath description from the HLSM state diagram
   - create 2 processes for behavioural controller description from the HLSM state diagram
   - create a process for behavioral output description from the HLSM state diagram.

Exam rules:

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02LQD Specification and Simulation of Digital Systems

2018/09/07

1. Combinational circuit design (5 points)

A combinational network with 4 inputs and 4 outputs is defined by the following truth table:

input	output
0000	0011
0001	0100
0010	0101
0011	0110
0100	0111
0101	1000
0110	1001
0111	1010
1000	1011
1001	1100
1010	1101
1011	1110
1100	1111
1101	0000
1110	0001
1111	0010

Design in VHDL a transcoder implementing the above table according to the following description styles:

1. dataflow using conditional signal assignments (WHEN ELSE statements)
2. dataflow using selected signal assignments (WITH SELECT WHEN statements)

2. FSM-D design (25 points)

The binomial coefficient is recursively defined as follows:

$$\binom{n}{k} = n!/(k! * \ (n-k)!)$$

The terminal cases are:

$$\binom{n}{n} = 1 \qquad \binom{n}{0} = 1$$

Iteratively it is computed as

$$\binom{n}{k} = n\ (n-1) \ldots .(n-k+1)/k!$$

Design a FSM-D to **iteratively** compute binomial coefficients. Inputs `n` and `k` ($n \geq k$) are pure binary numbers on 3 bits. It's up to the candidate to define the size of the `data_out` output. Computation starts when a *start* signal is asserted for one clock cycle. When computation is over the *ready* signal is asserted for one clock cycle. Reset is asynchronous. The output `data_out` makes sense only when *ready* is 1.

Steps:

1. Draw the HLSM state diagram and write the VHDL code to capture the HLSM behavior (**10 points**)
2. Design a FSM-D (**15 points**):
   - create 2 processes for behavioural datapath description from the HLSM state diagram
   - create 2 processes for behavioural controller description from the HLSM state diagram
   - create a process for behavioral output description from the HLSM state diagram.

Exam rules:

- upload on Portale della didattica in section Elaborati:
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02LQD Specification and Simulation of Digital Systems

2019/01/28

1. Combinational circuit design (3 points)

Binary coded decimal (BCD) encoding assigns a 4-bit binary code to each digit from 0 through 9 in base 10. The 4-bit BCD code for any particular single base-10 digit is its representation in binary notation. The 9-complement of a single BCD digit `d` is defined as 9-`d` by the following table :

BCD digit `d`	9-complement of `d`
0000	1001
0001	1000
0010	0111
0011	0110
0100	0101
0101	0100
0110	0011
0111	0010
1000	0001
1001	0000

Design in VHDL a combinational circuit implementing the above specification according to the following description styles:

- behavioural using IF THEN ELSIF statements
- behavioural using the CASE WHEN statement.

2. FSM design (7 points)

A Mealy machine acts as a recognizer of sequences transmitted on a serial line. It has 2 inputs and 2 outputs:

- the 1-bit input signal `En` enables machine operation when it is 1. Whenever it is 0 the machine returns to its initial state and waits for a new sequence
- the 1-bit input signal `X` is the serial line
- the 2-bit output signal $Z=Z_1Z_0$ is 10 when the input sequence 111 has been recognized on the last 3 bits, 01 when the input sequence 00 has been recognized on the last 2 bits, 00 otherwise.

Reset is synchronous.

Example

```
En   00111111111111110
X    01010001101001111
Z1   00000000000000110
Z0   00000110000010000
```

1. Draw the state diagram (**3 points**)
2. Design the FSM resorting to the VHDL behavioural description style with 2 processes (**4 points**).

3. FSM-D design (20 points)

Pell numbers are recursively defined as follows:

$P_n = 2P_{n-1} + P_{n-2}$ $\quad\quad$ n>1

$P_0 = 0$

$P_1 = 1$

The first 8 Pell numbers are thus: 0 1 2 5 12 29 70 169.

Design a FSM-D to **iteratively** compute Pell numbers of order `n`, where `n` is on 3 bits. Computation starts when a `start` signal is asserted for one clock cycle. When computation is over the `ready` signal is asserted for one clock cycle. Reset is asynchronous. The output is `P`. Define the appropriate size for `P`. The output `P` makes sense only when `ready` is 1.

Steps:

1. Draw the HLSM state diagram and write the VHDL code to capture the HLSM behavior (**7 points**)
2. Design a FSM-D (**13 points**):
   - create 2 processes for behavioural datapath description from the HLSM state diagram
   - create 2 processes for behavioural controller description from the HLSM state diagram
   - create a process for behavioral output description from the HLSM state diagram.

Exam rules:

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02LQD Specification and Simulation of Digital Systems

2019/02/11

1. Combinational circuit design (3 points)

In a base-3 numbering system, given 2 inputs `A` and `B`, a half-adder (no carry-in) computing the `Carry` (out) and the `Sum` is defined by the following table:

A	*B*	*Carry*	*Sum*
0	0	0	0
0	1	0	1
0	2	0	2
1	0	0	1
1	1	0	2
1	2	1	0
2	0	0	2
2	1	1	0
2	2	1	1

Encoding `A`, `B` and `Sum` on 2 bits a_1a_0, b_1b_0, s_1s_0:

- draw the truth table for the half-adder, defining proper output size
- design in VHDL the combinational circuit implementing the truth table according to the following description styles:
  3. dataflow using conditional signal assignments (WHEN ELSE statements)
  4. dataflow using selected signal assignments (WITH SELECT WHEN statements)

2. FSM design (12 points)

A **median filter** removes **lone 0s** in the input stream by changing each **lone 0** to a 1 on the output. A lone 0 is a 0 « sandwiched » between two 1s.

Example (lone 0s in boldface)
input stream: 01001**0**1**0**111**0**1001
output stream: 01001**1**1**1**111**1**1001

Note that the output stream is the (modified) input stream 2 two clock ticks later.

Design a Moore FSM with one serial input X and one serial output Z compliant with the specifications. Reset is asynchronous.

- Draw the state diagram (**7 points**)
- Design the FSM resorting to the VHDL behavioural description style with 2 processes (**5 points**).

3. FSM-D design (15 points)

Multiplication is a complex operation. A non-optimized algorithm is based on repetitive addition. For example 7 x 5 = 7 + 7 + 7 + 7 + 7. Design a FSM-D to perform multiplication of pure binary numbers by repeated addition. Inputs `a` and `b` are on 8 bits. Define a proper size for the output `z`. Computation starts when a `start` signal is asserted for one clock cycle. When computation is over the `ready` signal is asserted for one clock cycle. Reset is asynchronous. Values for output `z` are available at each computation step, but they make sense only when `ready` is 1.

Steps:

1. Draw the HLSM state diagram and write the VHDL code to capture the HLSM behavior (**5 points**)
2. Design a FSM-D (**10 points**):
   - create 2 processes for behavioural datapath description from the HLSM state diagram
   - create 2 processes for behavioural controller description from the HLSM state diagram.

Exam rules:

- upload on Portale della didattica in section Elaborati:
  - running VHDL code and testbench
  - short report (max 3 pages) describing the solution highlighting differences w.r.t. the solution handed in at the written exam

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02LQD Specification and Simulation of Digital Systems

2019/06/21

1. Combinational circuit design (5 points)

An unsigned number on 4 bits $n_3n_2n_1n_0$ is the input to a prime number detector circuit. The output z is 1 if and only if the input is a prime number. Design in VHDL a combinational circuit implementing the function above. Steps: write the truth table, the Karnaugh map, minimize the 2-level Boolean expressions, describe the circuit in VHDL resorting to the 2-level structural description style.

2. FSM design (12 points)

A synchronous sequential circuit has 3 inputs *Reset, Start* and *Stop* and 1 output *Run*. The duration of *Start* and *Stop* is not known a priori, i.e. it could span several clock cycles. When *Start* becomes 1, no matter the value of *Stop*, *Run* becomes 1. *Run* remains 1 until *Stop* becomes 1, no matter the value of *Start*, then *Run* goes to 0. Changes on *Run* are synchronous with the rising edge of the clock. Reset is asynchronous.

1. Draw the state diagram of the Moore machine (**3 points**)
2. Design the FSM resorting to the VHDL behavioural description style with 2 processes (**3 points**)
3. Design the FSM resorting to the VHDL behavioural description style with 1 process. The timing behavior may differ from the one of step 2 (**3 points**)
4. Design the FSM resorting to the VHDL behavioural description style with 1 process. The timing behavior must be the same of step 2. (**3 points**)

3. HLSM design (13 points)

Let X be a pure binary number on 2^n bits. Let L be its integer logarithm in base 2. Remember: L is the biggest integer such that $2^L \leq X$. Example: if $X = 312$, $L = \log_2(X) = 8$ as $2^8 \leq 312$ but $2^9 > 312$. Let *Start* be the input signal that starts computation when it is 1 and let *Done* be the output signal that tells the computation is over when it is 1. L is available at each clock tick, but it makes sense only when *Done* is 1.

Design a HLSM to compute the base-2 integer logarithm L of its input X when *Start* rises to 1 and n=4. *Done* set to 1 signals that a proper value of L is available on the output. Reset is synchronous. Draw the HLSM state diagram and write the VHDL code to capture the HLSM behavior.

Exam rules:

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    - running VHDL code and testbench
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02LQD Specification and Simulation of Digital Systems

2019/09/20

1. Combinational circuit design (4 + 2 + 2 points)

A car has a fuel level detector that outputs the current fuel level as a 3-bit binary number *lev*, where 000 means empty and 111 means full. Design a combinational circuit that illuminates a "low fuel" indicator light by setting output *L* to 1 when the fuel level drops below 3 (truth table, Karnaugh map, minimized Boolean expression) **(4 points)**. Describe the circuit in VHDL according to the following description styles:

- dataflow using conditional signal assignments (WHEN ELSE statements) **(2 points)**
- dataflow using selected signal assignments (WITH SELECT WHEN statements) **(2 points)**

2. FSM-D design (22 points)

Design an alarm system that sets a single output bit `alarm` to 1 at each clock tick when the average temperature of 4 consecutive samples exceeds a user-defined threshold value, otherwise it is set to 0. A16 bit unsigned input `CT` indicates the current temperature and a 16-bit unsigned input `WT` indicates the warning threshold. Reset is asynchronous.

Steps:

1. Draw the HLSM state diagram and write the VHDL code to capture the HLSM behavior (**10 points**)
2. Design a FSM-D (**12 points**):
   - create 2 processes for behavioural datapath description from the HLSM state diagram
   - create 2 processes for behavioural controller description from the HLSM state diagram.

Exam rules:

- upload on Portale della didattica in section Elaborati:
  - running VHDL code and testbench
  - short report (max 3 pages) describing the solution highlighting differences w.r.t. the solution handed in at the written exam

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02LQD Specification and Simulation of Digital Systems

2020/01/31

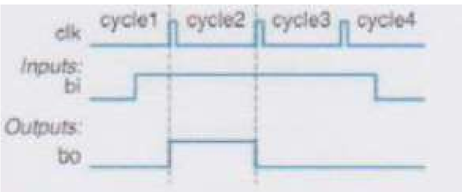
1. Combinational circuit design (3 points)

A network router connects multiple computers together and allows them to send messages to each other. If two or more computers send messages simultaneously, the messages collide and must be resent. Design a combinational collision detection circuit with 4 inputs `M0`, `M1`, `M2`, `M3` that are 1 when the corresponding computer is sending a message, 0 otherwise. The output `C` is 1 when a collision is detected, 0 otherwise.
Design in VHDL a combinational circuit implementing the above specification according to the following description styles:

- behavioural using IF THEN ELSIF statements
- behavioural using the CASE WHEN statement.

2. FSM design (12 points)

Design a Moore FSM that synchronizes a button press to a clock signal, such that, when the button is pressed (the 1-bit input signal `bi` is 1), the output signal `bo` is 1 for exactly one clock cycle. The `bo` synchronized output is used to prevent a single button press that lasts for multiple clock cycles from being interpreted as multiple button presses. The circuit waits for `bi` to return to 0 and then restarts operation waiting for `bi` to become 1 again, indicating that the button has been pressed once more. The following figure shows an example of the FSM's timing.



Reset is synchronous.

1. Draw the state diagram (**4 points**)
2. Design the FSM resorting to the VHDL behavioural description style with 2 processes (**4 points**).
3. Design the FSM resorting in Verilog with 2 always blocks (**4 points**).

3. FSM-D design (15 points)

Design a FSM-D for an alarm system. At each clock cycle the 1-bit output `alarm` is set to 1 whenever the average of the last 4 consecutive samples exceeds a threshold, otherwise it is set to 0. The first input is a 16-bit unsigned binary number `Temperature` for the physical quantity, e.g. the temperature. The second input is a 16-bit unsigned binary number `Threshold` for the warning threshold. Reset is asynchronous.
Steps:

1. Draw the HLSM state diagram and write the VHDL code to capture the HLSM behavior (**7 points**)
2. Design a FSM-D (**8 points**):
   - create 2 processes for behavioural datapath description from the HLSM state diagram
   - create 2 processes for behavioural controller description from the HLSM state diagram
   - connect datapath processes to controller processes to create a single architecture.

Exam rules:

- upload on Portale della didattica in section Elaborati a .zip archive including:
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  - short report (max 3 pages) describing the solution highlighting differences w.r.t. the solution handed in at the written exam

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02LQD Specification and Simulation of Digital Systems

2020/02/14

1. Combinational circuit design (3 points)

A combinational device has a 4-bit input signal `x`, a 1-bit input signal `sel`, a 3-bit output `y` and a 1-bit output `err`. `x` and `y` are unsigned binary numbers. The device computes the discrete base-2 logarithm of `x` and outputs it on `y`. When `sel`=0, `y` = $\lceil \log_2 x \rceil$, when `sel`=1, `y` = $\lfloor \log_2 x \rfloor$. Whenever `y` is not a valid result, `err` = 1.

Reminder: $\lceil R \rceil$ is the smallest integer S such that S≥ R. Examples: $\lceil 3.1 \rceil$=4, $\lceil 3.0 \rceil$=3.
Dually $\lfloor R \rfloor$ is the largest integer S such that S≤ R. Examples: $\lfloor 3.9 \rfloor$=3, $\lfloor 3.0 \rfloor$=3.

Design in VHDL a combinational circuit implementing the above specification according to the following description styles:

- behavioural using IF THEN ELSIF statements
- behavioural using the CASE WHEN statement.

2. FSM design (12 points)

A Moore FSM has a 1-bit input `X` and a 1-bit output `Z`. The FSM removes the first 1 from every sequence of 1s on it input and replaces it with a 0.

Example:
`X` = 001110100011110
`Z` = 00**0**110**0**000**0**1110

1. Draw the state diagram (**4 points**)
2. Design the FSM resorting to the VHDL behavioural description style with 2 processes (**4 points**).
3. Design the FSM resorting in Verilog with 2 always blocks (**4 points**).

Reset is asynchronous.

3. FSM-D design (15 points)

Design a FSM-D for a simple data encryption/decryption device. The device has 3 1-bit inputs `b`, `e` and `d` and a 16-bit signed input `I`. When `b`=1, no matter the values of `e` and `d`, the device stores the data on `I`. This data represents an offset value. When `e`=1 and `b`=0, no matter the value of `d`, the device encrypts the data it receives on input `I` adding to it the offset and outputs the result on a 16-bit signed output `O`. When `d`=1 and both `b`=0 and `e`=0, the device decrypts the data it receives on input `I` subtracting from it the offset and outputs the result on a 16-bit signed output `O`. Reset is asynchronous.

Steps:

1. Draw the HLSM state diagram and write the VHDL code to capture the HLSM behavior (**7 points**)
2. Design a FSM-D (**8 points**):
   - create 2 processes for behavioural datapath description from the HLSM state diagram
   - create 2 processes for behavioural controller description from the HLSM state diagram

Exam rules:

- upload on Portale della didattica in section Elaborati a .zip archive including:
  - running VHDL code and testbench
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02LQD Specification and Simulation of Digital Systems

2020/06/15

1. Combinational circuit design (3 points)

A car has a low-tire-pressure sensor that outputs the current tire pressure as a 5-bit binary number *P*. Create a circuit that illuminates a "low tire pressure" indicator light by setting the *T* output to 1 when the tire pressure *P* drops below 16.

Design in VHDL a combinational circuit implementing the above specification according to the following description styles:

- behavioural using IF THEN ELSIF statements
- behavioural using the CASE WHEN statement.

2. FSM design (12 points)

Build an electronic combination lock with a reset line *rst*, a serial input line *x* and an unlock output *u*. The unlock output *u* is at 1 if and only if the sequence 01011 has appeared on *x* in the last 5 clock ticks. The combination lock is a Moore machine with asynchronous reset.

- Draw the state diagram (**4 points**)
- Design the FSM resorting to the VHDL behavioural description style with 2 processes (**4 points**).
- Design the FSM resorting in Verilog with 2 always blocks (**4 points**).

3. FSM-D design (15 points)

Design a FSM-D to count the number of events on a 1-bit input line B. An event is any change of B from 0 to 1 or from 1 to 0. The number of events is output on a 16-bit output C. C rolls over (to 0) whenever its maximum value is reached. Reset is asynchronous.

Steps:

1. Draw the HLSM state diagram and write the VHDL code to capture the HLSM behavior (**7 points**)
2. Design a FSM-D (**8 points**):
   - create 2 processes for behavioural datapath description from the HLSM state diagram
   - create 2 processes for behavioural controller description from the HLSM state diagram
   - connect datapath processes to controller processes to create a single architecture.

Exam rules:

- upload on Portale della didattica in section Elaborati a .zip archive including:
  - running VHDL code and testbench
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02LQD Specification and Simulation of Digital Systems

2020/09/14

1. Combinational circuit design (3 points)

Design a 3 input ($X_2X_1X_0$), 1 output (Z) digital logic circuit which will take all the octal digits as its input and produce the even parity bit for the corresponding octal digit.

Design in VHDL a combinational circuit implementing the above specification according to the following description styles:

- behavioural using IF THEN ELSIF statements
- behavioural using the CASE WHEN statement.

2. FSM design (12 points

Design a garage door opener with a reset line *rst*, input lines *A*, *UP_LMT* and *DW_LMT* and output lines *MOT_UP* and *MOT_DW*. Input *A* starts the door to go up or down or stops the motion. It has thus 2 roles: it starts an action or, when an action is started, it stops it and starts the opposite one (e.g. if raising, stop raising and start lowering, or if lowering, stop lowering, start raising). Input *UP_LMT* indicates that maximum upward travel of the door has been reached. Input *DW_LMT* indicates that maximum downward travel of the door has been reached. Output *MOT_UP* causes the motor to run in direction to raise the door. Output *MOT_DW* causes the motor to run in direction to lower the door. Initially the door can be fully open or not. If not fully open, as soon as *A*=1 is detected, the door raises. If fully open, as soon as *A*=1 is detected, the door lowers. If fully closed, as soon as *A*=1 is detected, the door raises. Whenever *A*=0 is detected, motion stops and motor direction is inverted.

The FSM is a Moore machine with asynchronous reset.

- Draw the state diagram (**4 points**)
- Design the FSM resorting to the VHDL behavioural description style with 2 processes (**4 points**).
- Design the FSM resorting in Verilog with 2 always blocks (**4 points**).

3. FSM-D design (15 points)

Design an FSMD to control the speed of a conveyor belt. Speed is a 4 bit output *S* that is controlled by two buttons:

- the *Up* button increases speed by 1
- the *Down* button decreases speed by 1.

If both are pushed, no change in speed occurs. Speed must initialize to 0 upon startup. Speed must remain the range 0…15. Reset is asynchronous.

Steps:

1. Draw the HLSM state diagram and write the VHDL code to capture the HLSM behavior (**7 points**)
2. Design a FSM-D (**8 points**):
   - create 2 processes for behavioural datapath description from the HLSM state diagram
   - create 2 processes for behavioural controller description from the HLSM state diagram
   - connect datapath processes to controller processes to create a single architecture.

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02LQD Specification and Simulation of Digital Systems

2021/02/05

1. Combinational circuit design (2 points)

A museum has three rooms, each with a motion sensor (m0, m1, and m2) that outputs 1 when motion is detected. At night, the only person in the museum is one security guard who walks from room to room. Create a circuit that sounds an alarm (by setting an output A to 1) if motion is ever detected in more than one room at a time (i.e., in two or three rooms), meaning there must be one or more intruders in the museum.

Design in VHDL a combinational circuit implementing the above specification according to the following description styles:

- behavioural using IF THEN ELSIF statements
- behavioural using the CASE WHEN statement.

2. FSM design (16 points)

A Mealy FSM has a 1-bit input x and a 1-bit output z. Output z is 1 if and only if the most recent input, combined with the preceding three inputs, was not a valid BCD encoding of a decimal digit; otherwise, z = 0. Assume the BCD digits are received least significant bit first. Assume that in the initial state all previous inputs were 0. Reset is synchronous.

3. Draw the state diagram (**4 points**)
4. Design the FSM resorting to the VHDL behavioural description style with 2 processes (**4 points**).
5. Design the FSM resorting to the VHDL behavioural description style with 1 process (**4 points**).
6. Design the FSM resorting in Verilog with 2 always blocks (**4 points**).

3. HLSM design (12 points)

Jacobsthal numbers are recursively defined as follows:

$J_n = J_{n-1} + 2J_{n-2}$ $\qquad\qquad$ $n \geq 2$

$J_0 = 0$

$J_1 = 1$

The first 11 Jacobsthal numbers are thus: 0, 1, 1, 3, 5, 11, 21, 43, 85, 171, 341.

Design a HLSM to **iteratively** compute Jacobsthal numbers of order n, where n is a 4-bit input. Computation starts when a start signal is asserted for one clock cycle. When computation is over the ready signal is asserted for one clock cycle. Reset is asynchronous. The output is J. Define the appropriate size for J. The output J makes sense only when ready is 1.

Draw the HLSM state diagram (**6 points**) and write the VHDL code (**6 points**) to capture the HLSM behavior.

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02LQD Specification and Simulation of Digital Systems
2021/02/19

1. Combinational circuit design (2 points)

A museum has three rooms, each with a motion sensor (m0, m1, and m2) that outputs 1 when motion is detected. At night, the only person in the museum is one security guard who walks from room to room. Create a circuit that detects whether the guard is properly patrolling the museum, detected by *exactly* one motion sensor being 1 (by setting an output A to 1). If no motion sensor is 1, the guard may be sitting, sleeping, or absent.

Design in VHDL a combinational circuit implementing the above specification according to the following description styles:

- behavioural using IF THEN ELSIF statements
- behavioural using the CASE WHEN statement.

2. FSM design (16 points)

An FSM has a 1-bit input X and a 1-bit output Z. The output is the same as the input was three clock periods previously. The first three values of Z are 0.
Example:
X = 0 1 0 1 1 0 1 0 1 1 0 1 0 0 0 1
Z = 0 0 0 0 1 0 1 1 0 1 0 1 1 0 1 0

1. Draw the state diagram of the Mealy Machine (**4 points**)
2. Draw the state diagram of the Moore Machine (**4 points**)
3. Design the Moore FSM resorting to the VHDL behavioural description style with 2 processes (**4 points**).
4. Design the Mealy FSM resorting in Verilog with 2 always blocks (**4 points**).

3. HLSM (12 points)

A RAM has 1024 16-bit cells. A functional unit has a 16-bit input Inp, a 1-bit input Go, a 1-bit output Finish, an x-bit output Counter (x to be defined by the candidate) and the signals (to be defined by the candidate) required to access the RAM. The functional unit should read the RAM and output the number of cells whose value equals Inp. Operation starts when Go is asserted and Finish is set to 1 when operation is over and the result is available on Counter. Reset is synchronous. Design the functional unit as a HLSM, draw its state diagram (**6 points**) and write the VHDL code (**6 points**) to capture its behavior.

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02LQD Specification and Simulation of Digital Systems
2021/06/14

1. Combinational circuit design (2 points)

Design a combinational multiplier: input X is an integer in the range 0..5, output Z is Z = 3X. Define the number of input and output signals of the combinational circuit.

Design in VHDL a combinational circuit implementing the above specification according to the following description styles:

- behavioural using IF THEN ELSIF statements
- behavioural using the CASE WHEN statement.

2. FSM design (16 points)

An FSM has a 1-bit input X and a 1-bit output Z. It serves as a sequence detector that searches for a series of binary inputs to satisfy the pattern 01[0*]1, where [0*] is a sequence of any number of consecutive zeroes, including none. The output (Z) should become true every time the sequence is found.

Example:
X = 0 1 1 0 1 0 1 0 1 0 0 1
Z = 0 0 1 0 0 0 1 0 0 0 0 1

1. Draw the state diagram of the Mealy Machine (**4 points**)
2. Draw the state diagram of the Moore Machine (**4 points**)
3. Design the Moore FSM resorting to the VHDL behavioural description style with 2 processes (**4 points**).
4. Design the Mealy FSM resorting in Verilog with 2 always blocks (**4 points**).

3. HLSM (12 points)

A RAM has 1024 16-bit cells. A functional unit has a 1-bit input Go, a 1-bit output Finish, a 16-bit output Max, and the signals (to be defined by the candidate) required to access the RAM. The functional unit should read the RAM and output on Max the largest value. Operation starts when Go is asserted and Finish is set to 1 when processing is over and the result is available on Max. Reset is synchronous. Design the functional unit as a HLSM, draw its state diagram (**6 points**) and write the VHDL code (**6 points**) to capture its behavior.

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02LQD Specification and Simulation of Digital Systems
2021/09/03

1. Combinational circuit design (2 points)

Design a combinational converter from BCD to excess-3 code. Input X on 4 bits is a BCD digit. Output Z on 4 bits is the corresponding excess-3 code. The excess-3 code for a decimal digit is the result of adding 3 to the decimal digit.

Design in VHDL a combinational circuit implementing the above specification according to the following description styles:

- behavioural using IF THEN ELSIF statements
- behavioural using the CASE WHEN statement.

2. FSM design (12 points)

An FSM has a 1-bit input X and a 1-bit output Z. It serves as a sequential converter from BCD to excess-3 code. The BCD input arrives least significant bit (LSB) first. After 4 inputs the circuit resets to the initial state ready for another group of 4 inputs. The excess 3 code is output serially at the same time LSB first as well..

1. Draw the state diagram of the Mealy Machine (**4 points**)
2. Design the Mealy FSM resorting to the VHDL behavioural description style with 2 processes (**4 points**).
3. Design the Mealy FSM resorting in Verilog with 2 always blocks (**4 points**).

3. HLSM (16 points)

A system counts the number of events on a single-bit input B and always outputs that number unsigned on a 6-bit output C, which is initially 0. An event is a change from 0 to 1 or from 1 to 0. Assume that system count rolls over when the maximum value of C is reached. Reset is asynchronous.

1. Draw the state diagram of a HLSM that captures the above behavior (**6 points**)
2. write the corresponding VHDL code (**6 points**).
3. Design a FSM-D (**4 points**):
   - create 2 processes for behavioural datapath description from the HLSM state diagram
   - create 2 processes for behavioural controller description from the HLSM state diagram

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02LQD Specification and Simulation of Digital Systems

2022/01/26

1. Combinational circuit design (2 points)

Design a combinational circuit that receives a 4-bit unsigned binary number $X = x_3x_2x_1x_0$ and outputs a 4-bit unsigned binary number $Y = y_3y_2y_1y_0$ where $Y = (X + 5) \bmod 16$. Design in VHDL a combinational circuit implementing the above specification according to the following description styles:

- behavioural using IF THEN ELSIF statements
- behavioural using the CASE WHEN statement.

2. FSM design (12 points)

A serial combination lock allows entry to a locked room. It works as follows:

1. the user inputs the first combination and presses the ENTER button. The combination is recognized by a combinational network (whose design is not required) that outputs 1 if the combination is correct, 0 otherwise. This output is the COMB1 input to the system
2. the system waits for the user to enter the second combination and press the ENTER button. The combination is recognized by a combinational network (whose design is not required) that outputs 1 if the combination is correct, 0 otherwise. This output is the COMB2 input to the system
3. if both combinations are correct and have occurred in the correct sequence, the system unlocks the door (UNLOCK is set to 1) and waits to be reset. If any combination is wrong, the system doesn't unlock the door, it outputs 1 on the ERROR output and waits to be reset
4. reset is synchronous and brings the system back to the initial state.

Design a Moore FSM with 4 inputs RESET, COMB1, COMB2 and ENTER and 2 outputs UNLOCK and ERR exhibiting the above behavior. Assume the ENTER button when pressed generates a pulse for only one clock cycle.

1. Draw the state diagram (**4 points**)
2. Design the FSM resorting to the VHDL behavioural description style with 2 processes (**4 points**).
3. Design the FSM resorting in Verilog with 2 always blocks (**4 points**).

3. FSM-D design (16 points)

In **integer division** and **modulus**, the dividend is divided by the divisor into an integer quotient and a remainder. The integer quotient operation is referred to as integer division, and the integer remainder operation is the modulus. For example, if X = 93 and Y = 17, the integer quotient is 5 and the remainder is 8. A simple algorithm is based on repetitive subtraction. Design a FSM-D to perform integer division and modulus of pure binary numbers by repeated subtractions. Inputs `X` and `Y` are on 8 bits. Define a proper size for the integer quotient `q` and the remainder `r`. Computation starts when a `start` signal is asserted for one clock cycle. When computation is over the `ready` signal is asserted for one clock cycle. Reset is asynchronous. Values for outputs `q` and `r` are available at each computation step, but they make sense only when `ready` is 1.

Steps:

1. Draw the HLSM state diagram and write the VHDL code to capture the HLSM behavior (**6 points**)

2. Design a FSM-D (**10 points**):

- create 2 processes for behavioural datapath description from the HLSM state diagram
- create 2 processes for behavioural controller description from the HLSM state diagram.

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02LQD Specification and Simulation of Digital Systems
2022/02/11

1. Combinational circuit design (2 points)

A combinational circuit has a 4-bit input $M=m_0m_1m_2m_3$ and it generates a 1-bit output Z according to the following rules:

- Z is don't care if all the 4 bits of M are either at 0 or at 1
- Z is 1 if an odd number of bits of M is at 1
- Z is 0 if an even number of bits of M is at 1.

Design in VHDL a combinational circuit implementing the above specification according to the following description styles:

- behavioural using IF THEN ELSIF statements
- behavioural using the CASE WHEN statement.

2. FSM design (16 points)

An FSM has a 1-bit input X and a 1-bit output Z. Its goal is to remove the first 1 on every string of 1s on the input.

Example:

X = 0 1 0 1 1 0 1 0 1 1 1 1 0 0 0 1
Z = 0 0 0 0 1 0 0 0 0 1 1 1 0 0 0 0

1. Draw the state diagram of the Moore Machine (**4 points**)
2. Draw the state diagram of the Mealy Machine (**4 points**)
3. Design the **Moore** FSM resorting to the **VHDL** behavioural description style with 2 processes (**4 points**).
4. Design the **Mealy** FSM resorting in **Verilog** with 2 always blocks (**4 points**).

3. HLSM (12 points)

A RAM has 1024 16-bit cells. A functional unit has a 10-bit input Addr, a 1-bit input Go and a 1-bit output Finish and the signals (to be defined by the candidate) required to access the RAM. When the Go signal is asserted for 1 clock cycle, the functional unit reads a first memory address on its Addr input. When the Go signal is asserted a second time for 1 clock cycle, the functional unit reads a second memory address on its Addr input. It then sets to 0 all RAM cells whose addresses are in the range from the first address read to the second address read, boundaries included. Assume that the inputs are correct, e.g. the second address is always no less than the first one. Output Finish is set to 1 when operation is over. Reset is synchronous. Design the functional unit as a HLSM, draw its state diagram (**6 points**) and write the VHDL code (**6 points**) to capture its behavior.

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02LQD Specification and Simulation of Digital Systems
2022/06/27

1. Combinational circuit design (2 points)

Let X be the 1-bit Minuend, let Y be the 1-bit Subtrahend, let B_{in} be the 1-bit Borrow-in. X, Y and B_{in} are inputs. Let D be the 1-bit Difference, let B_{out} be the 1-bit Borrow-out. D and B_{out} are outputs. Design in VHDL a **full-subtractor** according to the following description styles:

- behavioural using IF THEN ELSIF statements
- behavioural using the CASE WHEN statement.

2. FSM design (16 points)

An FSM has a 1-bit input X and a 2-bit output Z. Its goal is to count how many 01 sequences appear in a 4-bit window. This value is available on Z once the 4th bit of the input sequence has been read, otherwise Z is at 0.

Example:
$X =$ 0 1 0 1 1 0 1 0 1 1 1 1 0 0 0 1
$Z_1 =$ 0 0 0 1 0 0 0 0 0 0 0 0 0 0 0 0
$Z_0 =$ 0 0 0 0 0 0 0 1 0 0 0 0 0 0 0 1

1. Draw the state diagram of the Moore Machine (**4 points**)
2. Draw the state diagram of the Mealy Machine (**4 points**)
3. Design the **Moore** FSM resorting to the **VHDL** behavioural description style with 2 processes (**4 points**).
4. Design the **Mealy** FSM resorting in **Verilog** with 2 always blocks (**4 points**).

3. HLSM (12 points)

A RAM has 1024 16-bit cells. A functional unit has a 10-bit input Addr and a 1-bit input Go as well as the signals (to be defined by the candidate) required to access the RAM. When the Go signal is asserted for 1 clock cycle, the functional unit reads a memory address on its Addr input, then it counts how many bits are at 1 in the corresponding 16-bit cell in the RAM (*popcount*). Reset is synchronous. Design the functional unit as a HLSM, draw its state diagram (**6 points**) and write the VHDL code (**6 points**) to capture its behavior.

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02LQD Specification and Simulation of Digital Systems
2022/09/16

1. Combinational circuit design (2 points)

Let X and Y be two BCD digits. Design in VHDL a **1-digit BCD adder** according to the following description styles:

- behavioural using IF THEN ELSIF statements
- behavioural using the CASE WHEN statement.

It is up to the candidate to define the output signals and their size.

2. FSM design (16 points)

Design a FSM that serves as a controller for an elevator. The elevator can be at one of 3 floors: Ground, First or Second. There is one button that controls the elevator, and it has two values: Up or Down. Also, there are 3 lights in the elevator that indicate the current floor: Red for Ground, Green for First, Blue for Second. At each time step, the controller checks the current floor and current input, changes floors and lights in the obvious way.

1. Draw the state diagram of the Moore Machine (**4 points**)
2. Design the **Moore** FSM resorting to the **VHDL** behavioural description style with 2 processes (**4 points**).
3. Design the **Moore** FSM resorting to the **VHDL** behavioural description style with 1 process (**4 points**)
4. Design the **Moore** FSM resorting in **Verilog** with 2 always blocks (**4 points**).

3. HLSM (12 points)

A RAM has 1024 16-bit cells. A functional unit has a 10-bit input `Addr` and a 1-bit input `Go` as well as the signals (to be defined by the candidate) required to access the RAM. When the `Go` signal is asserted for 1 clock cycle, the functional unit reads a memory address on its `Addr` input, then it computes the difference (positive or negative) between the number of bits at 1 and at 0 in the corresponding 16-bit cell in the RAM (*diffcount*). Reset is synchronous. Design the functional unit as a HLSM, draw its state diagram (**6 points**) and write the VHDL code (**6 points**) to capture its behavior.

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02LQD Specification and Simulation of Digital Systems

2023/01/23

1. Combinational circuit design (2 points)

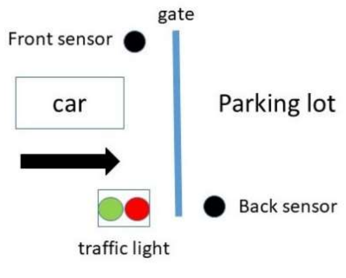
Design a combinational circuit that serves as a nonlinear LUT. It receives an 8-bit unsigned binary number LUTin and outputs an 8-bit unsigned binary number LUTout. The 2 nibbles of LUTout depend on the 2 nibbles of LUTin according to the following table:

Nibble #1 (bits 7:4)		Nibble #2 (bits 3:0)	
LUTin(7:4)	LUTout(7:4)	LUTin(3:0)	LUTout(3:0)
0000	0001	0000	1111
0001	1011	0001	0000
0010	1001	0010	1101
0011	1100	0011	0111
0100	1101	0100	1011
0101	0110	0101	1110
0110	1111	0110	0101
0111	0011	0111	1010
1000	1110	1000	1001
1001	1000	1001	0010
1010	0111	1010	1100
1011	0100	1011	0001
1100	1010	1100	0011
1101	0010	1101	0100
1110	0101	1110	1000
1111	0000	1111	0110

Design in VHDL a combinational circuit implementing the above specification according to the behavioral description style using the CASE WHEN statement (it is not necessary to explicitly write all input combinations, what matters is the overall structure of the description).

2. FSM design (12 points)

The entrance gate of a parking lot is controlled by a car parking system. Just in front of the gate there is a front sensor F and a traffic light (TL). Just after the gate there is a back sensor B. An external device (not to be designed) checks the code of the car and generates an ok input to the system.



The traffic light may be off ($TL=l_1l_0=00$) or flashes green ($TL=l_1l_0=01$) or red ($TL=l_1l_0=10$). The front sensor either detects a car willing to enter the parking lot (F=1) or no car is willing to enter (F=0). The back sensor detects whether the car has crossed the gate and is in the parking lot (B=1) or still crossing (B=0). The car never backs up.

If no car is trying to enter the parking lot (F=0), the system keeps the traffic light off and the gate is closed. If the front sensor detects a vehicle (F=1), the system turns on the red light as long as the code is wrong (ok=0). As soon as a correct code is detected (ok=1), the systems flashes the green light and opens the gate. While the car is crossing the gate, the back sensor is B=0 and no other car willing to enter is allowed to (the traffic light is red). As soon as the car has crossed the gate, B=1 and operations continue depending on the presence/absence of a car willing to enter.

Design a **Mealy** FSM with F, B and ok as inputs and TL as output.
1. Draw the state diagram (**4 points**)
2. Design the FSM resorting to the VHDL behavioural description style with 2 processes (**4 points**).
3. Design the FSM resorting in Verilog with 2 always blocks (**4 points**).

3. FSM-D design (16 points)

In arithmetic the **least common multiple** of two positive integers *a* and *b*, usually denoted by **lcm(*a*, *b*)**, is the smallest positive integer that is divisible by both *a* and *b*. Since division of integers by zero is undefined, this definition has meaning only if *a* and *b* are both different from zero.

Examples:

- a = 0, b ≠ 0: lcm(0, b) = 0
- a ≠ 0, b = 0: lcm(a, 0) = 0
- a = 7, b = 7: lcm(7, 7) = 7
- a = 12, b = 18: lcm(12, 18) = 36

A simple algorithm to compute lcm(a,b) is:

```
m=a
n=b
while (m != n) {
  if (m < n)
    m=m+a
  else
    n=n+b
}
lcm=m
```

Design a FSM-D to compute the lcm of 2 pure binary numbers a and b on 8 bits implementing the above algorithm. Computation starts when a start signal is asserted for one clock cycle. When computation is over the ready signal is asserted for one clock cycle. Reset is asynchronous. Values for lcm are available at each computation step, but they make sense only when ready is 1.

Steps:

4. Define the appropriate size for the lcm output (**2 points**)

5. Draw the HLSM state diagram and write the VHDL code to capture the HLSM behavior (**6 points**)

6. Design a FSM-D:
- create 2 processes for behavioural datapath description from the HLSM state diagram (4 **points**)
- create 2 processes for behavioural controller description from the HLSM state diagram (**4 points**).

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02LQD Specification and Simulation of Digital Systems

2023/02/09

1. Combinational circuit design (2 points)

Design a four-input *priority arbiter*, a circuit which accepts four inputs and outputs the index of the input with the highest value. Inputs are pure binary numbers on 8 bits. In the event of a tie, the circuit outputs the lowest index that has the highest value.

Example:

- suppose the four inputs are 28, 32, 47, and 19. The arbiter will output 2 because input 2 has the highest value, 47
- suppose the four inputs are 17, 23, 19, 23, the arbiter will output 1 because, of the two inputs (1 and 3) with the high value of 23, input 1 has the lowest index.

2. FSM design (12 points)

Design a sequential parity detector. Data arrives at a single input X, with one new bit per clock cycle. The data is grouped into packets of four bits, where the fourth bit is a parity bit. The system checks even parity: if there is an odd number of 1s in the first three bits, the fourth bit must be a 1. If an incorrect parity bit is detected, an error signal ERR is asserted. Reset is synchronous.

Design a **Mealy** FSM with X as input and ERR as output.

1. Draw the state diagram (**4 points**)
2. Design the FSM resorting to the VHDL behavioural description style with **1 process** (**4 points**).
3. Design the FSM resorting in Verilog with **2 always blocks** (**4 points**).

3. FSM-D design (16 points)

Population count, also known as popcount or Hamming weight, is the number of 1s in a data word. Design an FSM-D to compute the popcount of 8 bit data words din. When a new data word is available, the dinValid input signal is asserted and popcount computation starts. No more data words are taken into account until computation is over and the dinReady output signal is negated. When computation is over, the popcount is output on the popCnt output, the popCntValid output signal is asserted and the FSM-D signals it is ready to accept a new data word by asserting the dinReady output signal. Reset is synchronous.

1. Define the appropriate size for the popCnt output (**2 points**)
2. Draw the HLSM state diagram and write the VHDL code to capture the HLSM behavior (**6 points**)
3. Design a FSM-D:
   - create 2 processes for behavioural datapath description from the HLSM state diagram (**4 points**)
   - create 2 processes for behavioural controller description from the HLSM state diagram (**4 points**).

Exam rules:

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02LQD Specification and Simulation of Digital Systems

2023/04/13

1. Combinational circuit design (2 points)

Thermometer-to-binary (TM2B) encoders are important components of Flash Analog-to-Digital Converters (FADC). The thermometer outputs a pure binary signal X on 7 bits. The encoder receives X and generates Z on 3 bits according to the following table:

X	Z
0000000	000
0000001	001
0000011	010
0000111	011
0001111	100
0011111	101
0111111	110
1111111	111

Design in VHDL an encoder implementing the above specification according to the following description styles:

3. behavioural using IF THEN ELSEIF statements
4. behavioural using the CASE WHEN statement.

2. FSM design (12 points)

A Mealy machine acts as a recognizer of sequences transmitted on a serial line. It has 2 inputs and 2 outputs:

- the 1-bit input signal `En` enables machine operation when it is 1. Whenever it is 0 the machine returns to its initial state and waits for a new sequence
- the 1-bit input signal `X` is the serial line
- the 2-bit output signal $Z=Z_1Z_0$ is 10 when the input sequence 111 has been recognized on the last 3 bits, 01 when the input sequence 00 has been recognized on the last 2 bits, 00 otherwise.

Reset is synchronous.

<u>Example</u>

```
En    0011111111111111110
X     0101000110100111111
Z1    0000000000000000110
Z0    0000011000000100000
```

Design a **Mealy** FSM with the above specified inputs, outputs and behavior:

1. Draw the state diagram (**4 points**)
2. Design the FSM resorting to the VHDL behavioural description style with **1 process** with the same timing behavious as the 2-process description (**4 points**).
3. Design the FSM resorting in Verilog with **2 always blocks** (**4 points**).

3. FSM-D design (16 points)

Multiplication is a complex operation. A non-optimized algorithm is based on repetitive addition. For example -7 x 5 = -(7 + 7 + 7 + 7 + 7). Design a FSM-D to perform multiplication of signed binary numbers by repeated addition. Inputs `a` and `b` are signed binary numbers on 8 bits. Define a proper size for the output `z`. Computation starts when a `start` signal is asserted for one clock cycle. When computation is over the `ready` signal is asserted for one clock cycle. Reset is asynchronous. Values for output `z` are available at each computation step, but they make sense only when `ready` is 1.

Steps:

1. **1.** Define a proper size for the output z (**2 points**)
2. **2.** Draw the HLSM state diagram and write the VHDL code to capture the HLSM behavior (**6 points**)
3. **3.** Design a FSM-D:
   - create 2 processes for behavioural datapath description from the HLSM state diagram (**4 points**)
   - create 2 processes for behavioural controller description from the HLSM state diagram. (**4 points**).

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02LQD Specification and Simulation of Digital Systems

2023/06/21

1. Combinational circuit design (2 points)

8-4-2-1 is a self-complementary, biased and redundant code. Each decimal digit is expressed on 4 bits. The following table matches BCD and 8-4-2-1:

BCD	8 4 -2 -1
0000	0000
0001	0111
0010	0110
0011	0101
0100	0100
0101	1011
0110	1010
0111	1001
1000	1000
1001	1111

Design in VHDL a BCD to 8-4-2-1 transcoder implementing the above table according to the following description styles:

- dataflow using conditional signal assignments (WHEN ELSE statements)
- dataflow using selected signal assignments (WITH SELECT WHEN statements).

2. FSM design (12 points)

In digital communication, a special synchronization pattern, known as a ***preamble,*** is used to indicate the beginning of a packet. For example, the Ethernet II preamble includes eight repeating octets of "10101010". Design a synchronous circuit that generates the "10101010" pattern. The circuit has a 1 bit input signal, `start`, and a 1 bit output signal, `data_out`. When `start` is 'l', the "10101010" output will be generated in the next eight clock cycles. Reset is synchronous.

Design a **Mealy** FSM with the above specified inputs, outputs and behavior:

- Draw the state diagram (**4 points**)
- Design the FSM resorting to the VHDL behavioural description style with **1 process** with the same timing behaviour as the 2-process description (**4 points**).
- Design the FSM resorting in Verilog with **2 always blocks** (**4 points**).

3. FSM-D design (16 points)

Pell numbers are recursively defined as follows:

$P_n = 2P_{n-1} + P_{n-2}$ n>1

$P_0 = 0$

$P_1 = 1$

The first 8 Pell numbers are thus: 0 1 2 5 12 29 70 169.

Design a FSM-D to **iteratively** compute Pell numbers of order `n`, where `n` is on 3 bits. Computation starts when a `start` signal is asserted for one clock cycle. When computation is over the `ready` signal is asserted for one clock cycle. Reset is asynchronous. The output is `P`. Define the appropriate size for `P`. The output `P` makes sense only when `ready` is 1.

Steps:

3. Define a proper size for the output `P` (**2 points**)
4. Draw the HLSM state diagram and write the VHDL code to capture the HLSM behavior (**6 points**)

5. Design a FSM-D:

- create 2 processes for behavioural datapath description from the HLSM state diagram (**4 points**).
- create 2 processes for behavioural controller description from the HLSM state diagram (**4 points**).

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02LQD Specification and Simulation of Digital Systems

2023/09/21

1. Combinational circuit design (2 points)

Thermometer-to-binary (TM2B) encoders are important components of Flash Analog-to-Digital Converters (FADC). The thermometer outputs a pure binary signal X on 7 bits. The encoder receives X and generates Z on 3 bits according to the following table:

X	Z
0000000	000
0000001	001
0000011	010
0000111	011
0001111	100
0011111	101
0111111	110
1111111	111

Design in VHDL an encoder implementing the above specification according to the following description styles:

- behavioural using IF THEN ELSEIF statements
- behavioural using the CASE WHEN statement.

2. FSM design (12 points)

A **median filter** removes **lone 1s** in the input stream by changing each **lone 1** to a 0 on the output. A lone 1 is a 1 « sandwiched » between two 0s.

Example (lone 1s in boldface)
input stream: 10110**1**01110**1**0**1**01
output stream: 10110**0**01110**0**0**0**01

Note that the output stream is the (modified) input stream 2 two clock ticks later.

Design a Moore FSM with one serial input X and one serial output Z compliant with the specifications. Reset is asynchronous.
Design a **Moore** FSM with the above specified inputs, outputs and behavior:

- Draw the state diagram (**4 points**)
- Design the FSM resorting to the VHDL behavioural description style with **1 process** with the same timing behaviour as the 2-process description (**4 points**).
- Design the FSM resorting in Verilog with **2 always blocks** (**4 points**).

4. FSM-D design (16 points)

Tribonacci numbers are recursively defined as follows:

$TRIB_n = TRIB_{n-3} + TRIB_{n-2} + TRIB_{n-1}$ $\qquad n \geq 3$
$TRIB_0 = 0$
$TRIB_1 = 1$
$TRIB_2 = 1$

The first 10 Tribonacci numbers are thus: 0 1 1 2 4 7 13 24 44 81.

Design a FSM-D to **iteratively** compute Tribonacci numbers of order `n`, where `n` is on 4 bits. Computation starts when a `start` signal is asserted for one clock cycle. When computation is over the `ready` signal is asserted for one clock cycle. Reset is asynchronous. The output is `P`. Define the appropriate size for `P`. The output `P` makes sense only when `ready` is 1.

Steps:

1. Define a proper size for the output `P` (**2 points**)
2. Draw the HLSM state diagram and write the VHDL code to capture the HLSM behavior (**6 points**)
3. Design a FSM-D:
   - create 2 processes for behavioural datapath description from the HLSM state diagram (**4 points**).
   - create 2 processes for behavioural controller description from the HLSM state diagram (**4 points**).

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